Modern Signal Analysis and Data Processing

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Question 1 Briefly explain the Laplace transform and compare it with the Fourier transform from the perspective of convergence.

The Laplace transform can be regarded as the Fourier transform of the signal multiplied by a convergence factor $e^{-\sigma t}$. Its definition is:

$$\mathcal{L}\{f(t)\} = \int_0^{+\infty} f(t)e^{-st}dt, \quad s = \sigma + j\omega$$

For many unbounded or growing signals, the standard Fourier integral fails to converge. By multiplying the original function by $e^{-st} = e^{-\sigma t}e^{-j\omega t}$, the real exponential term $e^{-\sigma t}$ attenuates the signal amplitude, forcing the integral to converge.

Similarities: Both are linear integral transforms that convert time-domain convolution into frequency-domain multiplication.

Differences: Fourier transform only works for absolutely integrable signals. In contrast, the Laplace transform extends the Fourier transform by introducing the convergence factor $e^{-\sigma t}$, which supports non-decaying and divergent signals and is more suitable for system transient and stability analysis.

Question 2 Prove the Laplace convolution property: $\mathcal{L}\{f * g\} = \mathcal{L}\{f\}\mathcal{L}\{g\}$.

Proof: The convolution of two causal functions is defined as:

$$(f * g)(t) = \int_0^t f(\tau)g(t-\tau)d\tau$$

Taking the Laplace transform:

$$\mathcal{L}\{f * g\} = \int_0^\infty \left(\int_0^t f(\tau)g(t-\tau)d\tau \right) e^{-st}dt$$

Swap the integration order and substitute $u = t - \tau$:

$$\begin{aligned} \mathcal{L}\{f * g\} &= \int_0^\infty f(\tau)e^{-s\tau}d\tau \int_0^\infty g(u)e^{-su}du \\ &= \mathcal{L}\{f(t)\} \cdot \mathcal{L}\{g(t)\} \end{aligned}$$

The convolution property is proven.